itFits Report

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Specification of Products and Services

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Session 4: Technical Drawings, Exploded View, and BOM

1. PRODUCT ARCHITECTURE

The product architecture of **itFits** is designed as a high-performance, modular system that balances ergonomic support with extreme customization. The system is built around three core components that integrate to create a "perfect fit" while allowing users to swap modules based on their activity.

The Three Core Components

The foundation of the architecture consists of the following parts:

- **Ergonomic Back Sleeve (Back Belt):** This is a 9cm wide section made from **"vintage" leather**. Its primary function is to provide lumbar support and ensure the belt molds to the user's figure, preventing the "bounce" common in traditional fanny packs.
- **Main Adjustable Strap:** A 5cm wide high-resistance strap made from **100% recycled nylon or polyester thread**. This strap handles the structural tension and serves as the mounting rail for all modules.
- **UNBRK Quick-Release Buckle:** A 6cm x 9cm two-piece hardware set made of **injection-molded polycarbonate**. It is designed for "rapid deployment," allowing the user to open and close the belt efficiently.

Integration: How the Parts Work Together

The assembly of the belt is designed to be intuitive and adjustable:

- **Threading the System:** The Back Sleeve features **belt loops**. The Main Strap is threaded through these loops, which allows the ergonomic back part to be positioned precisely.

- **Buckle Attachment:** The ends of the nylon strap are joined to the UNBRK buckle via **industrial stitching**, ensuring a permanent and secure bond that can withstand high-demand environments.
- **Fit Adjustment:** Because the strap slides through the back sleeve's loops, the user can adjust the tension and position of the support padding regardless of their body type, achieving a **"tailor-made" system**.

The Modular System (Mini Bags)

- **Module Variety:** Users can choose from various "Mini Bag" styles, such as the **Zippo phone bag**, bottle carriers, or labial carriers.
- **The "Slip" Attachment:** Each mini bag now features a **slip or slot on its back panel**.
- **Mounting Logic:** To attach a module, the user simply slides the **nylon strap through the slip** on the back of the mini bag. This creates a secure, **bounce-free join**, allowing modules to be added, removed, or repositioned along the length of the strap.

Operational Synergy

When worn, these parts work in unison to deliver the **"1 is all I need"** experience. The Back Sleeve distributes weight and protects the skin from the strap's tension; the strap provides the "infinite possibilities" for module placement; and the "slip" mechanism ensures that whether the user is at a music festival or a construction site, their gear stays exactly where they put it.

The assembly sequence for the **itFits** modular utility belt follows a precise order of industrial fabrication and manual integration, transitioning from raw materials to a fully customized wearable system.

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2. PARTS BREAKDOWN

Manufactured Parts (Custom Pieces)

These are the core components specifically designed and fabricated for the project to achieve its unique modularity and ergonomic fit.

- **UNBRK Quick-Release Buckle:** A custom-engineered, three-piece hardware set (6cm x 9cm) made from high-strength polycarbonate through **plastic injection molding**.
- **Ergonomic Back Sleeve:** A custom-shaped 9cm wide lumbar support piece constructed from "vintage" leather or pleather, featuring internal belt loops to house the main strap.
- **Modular Mini Bags:** Custom-designed storage units in various sizes and materials, including:
    - **Zippo Phone Bag:** A rigid-shell module (109g).
    - **Bottle Mini Bags:** Available in standard (100g) and "extreme" versions (88g).
    - **Labial Carrier:** An "extreme version" modular attachment.

Commercial Parts (Purchased Components)

These are raw materials or off-the-shelf components sourced from existing suppliers to be integrated into the final assembly.

- **Recycled Textile Straps:** 5cm wide webbing woven from **100% recycled nylon or polyester thread**, sourced from partners like Sutexsa or Deportextil.
- **Raw Leather Scraps:** Cowhide or "vintage" leather collected from other manufacturers' waste to promote sustainability.
- **Polycarbonate Pellets:** Raw plastic material used for the injection molding of the UNBRK buckle and rigid modules.
- **Industrial Hardware:** Standard components, such as **buttons** used to tighten the ergonomic strap structure.

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Consumables + Finishes

Materials used during the production process to finalize the product's aesthetic and structural integrity.

- **Industrial Stitching Thread:** High-resistance thread used for the reinforced assembly of textile and leather components.
- **Dyes and Printed Finishes:** High-saturation dyes and printing processes used to achieve "striking colors" and patterns like the **zebra motif**.
- **Prototyping Filaments:** 400g of filament (approx. $100 MXN) used for 3D printing initial structural validations.
- **Packaging and Logistics Materials:** Shipping materials required for the **"Give Back for Store Credit"** program, enabling users to mail in old modules for internal recycling.

3. EXPLODED VIEW

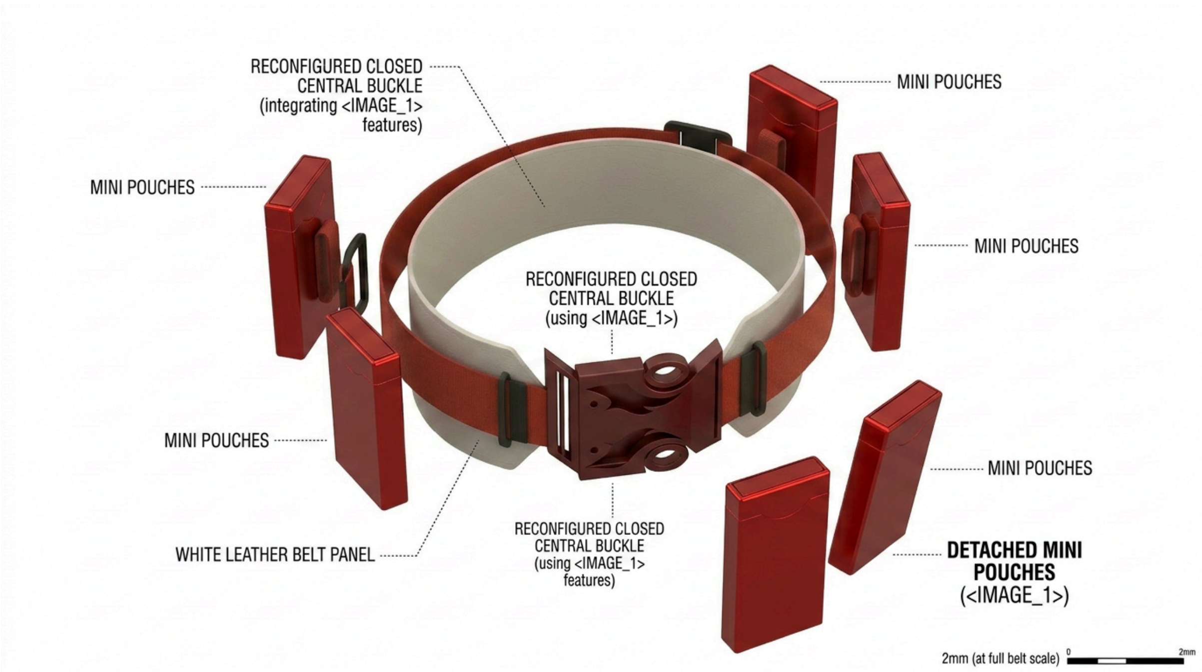

Fig. 4.1 Exploded view made with VIZCOM

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4. DRAWING CHECK LIST

For the final technical report of the **itFits** belt, the sources indicate that several specific drawings and technical breakdowns are required to document the engineering specifications, assembly logic, and modularity of the system.

The following drawings are essential for the technical report:

Core Belt Assembly & Components

- **Main Assembly Drawing:** An overall technical drawing with dimensions for the entire belt system, including total length (e.g., 670mm) and specific widths for the different sections.
- **Ergonomic Back Sleeve:** A detailed breakdown of the **9cm wide** lumbar support part, documenting the internal integration of memory foam and the placement of belt loops.
- **Main Adjustable Strap:** Technical specifications for the **5cm wide** high-resistance strap, including its weaving pattern and the stitching points where it joins the buckle.
- **Belt Loops & Spacing:** Specific drawings detailing the dimensions (e.g., 70mm height, 10mm spacing) and frequency of the loops on the back sleeve used to thread the main strap.

Hardware & Mechanical Parts

- **UNBRK Quick-Release Buckle:** A technical breakdown of the **two-piece (6cm x 9cm)** or three-piece injection-molded hardware. This should include individual drawings for each part to facilitate the **polycarbonate injection molding** process.
- **Buckle Joining Structures:** Drawings of the **button-like structures** used to tighten the buckle assembly altogether.

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Modular Attachments (Mini Bags)

The report must include technical drawings for the various "mini bag" styles, documenting their specific dimensions and purposes:

- **Zippo Phone Bag:** Drawings of the rigid or fabric module designed for smartphones.
- **Bottle Carrier Modules:** Specifications for both the standard and "extreme version" bottle straps.
- **Labial Carrier (Extreme Version):** Detailed views of the lid and carrier body for this specialized module.
- **Labial Carrier (normal version):** Detailed views of the lid and carrier body for this specialized module.
- **Attachment Slip Mechanism:** A specific drawing of the **slip or slot on the back panel** of each mini bag, showing how the 5cm nylon strap slides through it to create a bounce-free join.

Assembly & Integration Drawings

- **Exploded View:** A drawing showing how the three core components (Back Sleeve, Main Strap, and UNBRK Buckle) fit together.
- **Mounting Logic Diagram:** A visual guide illustrating the **"slip" attachment** method, where the strap threads through the modules, replacing the older hexagonal system.

Wearing Versatility: Sketches or photos showing the belt positioned in various wearable ways (waist, hips, or cross-body/chest) to validate its ergonomic and functional range.

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5. PRELIMINARY BOM LIST

Based on the preliminary Bill of Materials (BOM) and the supplier documentation provided in the sources, the following table completes the BOM with the required technical processes, partners, and known costs.

Part / Component	Material	Process	Supplier / Partner	Estimated Cost (MXN)
Main Adjustable Strap	Nylon thread (5cm wide)	Industrial weaving and stitching	**Sutexsa / Deportextil**	**$279** (2" x 3.3 yards)
Ergonomic Back Sleeve	Cow hide / "Vintage" leather scraps	Industrial-grade leather stitching	**Lefarc**	**Depends on the leather type and measure**
UNBRK Buckle (2 parts)	High-strength Polycarbonate	Plastic injection molding	**UNBROKEN**	To be defined*
Phone Bag (Zippo - 109g)	Nylon or Polycarbonate	Injection molding	**UNBROKEN**	To be defined
Bottle Mini Bag (100g/88g)	Nylon or Polycarbonate	Injection molding	**UNBROKEN**	To be defined
Assembly Button	Industrial hardware	Manual/Mechanical fastening	**Botones Loren**	$30 a piece

Note: While the final industrial manufacturing cost for injection-molded parts is not yet defined, the sources note a cost of **$100 MXN for 400g of filaments used in prototype development.*

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6. ASSEMBLY SEQUENCE

Phase 1: Component Fabrication

The process begins with the independent manufacturing of the belt's three primary structural elements:

1. **Ergonomic Back Sleeve Construction:**
   - **Cutting:** Raw "vintage" leather or high-quality pleather is cut to size, specifically for a **9cm wide** back sleeve section.
   - **Loops:** Internal **belt loops or slots** are integrated into the sleeve using industrial stitching, allowing it to house the main strap.
2. **Main Strap Production:**
   - High-resistance **recycled nylon or polyester thread** is woven into **5cm wide** straps. These straps handle the structural tension and serve as the mounting rail for all modules.
3. **Hardware Injection Molding:**
   - The **UNBRK quick-release buckle** (6cm x 9cm) and various rigid modular components are produced via **polycarbonate plastic injection molding**.

Phase 2: Sub-Assembly and Integration

Once the individual components are ready, they are joined into a core functional unit:

1. **Buckle Attachment:** The ends of the nylon strap are permanently secured to one end of the UNBRK buckle using **reinforced industrial stitching**.
2. **Threading the System:** The finished main strap is **threaded through the belt loops** of the ergonomic back sleeve. Because the strap slides freely through these loops, the back can be positioned precisely according to the user's body type.

Phase 3: Customization (The Modular System)

The final configuration is determined by the user's specific needs:

1. **Module Selection:** The user chooses from various "Mini Bag" styles, such as the **Zippo phone bag**, bottle carriers, or labial carriers.
2. **Slip Attachment:** Each mini bag features a **slip or slot on its back panel**.
3. **Mounting Logic:** To attach a module, the user simply **slides the nylon strap through the slip** on the back of the bag. This creates a secure, bounce-free join that allows modules to be added, removed, or repositioned along the strap at any time.

Phase 4: Final Fit and Wearing

1. **Adjustability:** The user adjusts the tension of the main strap to achieve a **"tailor-made" fit**.
2. **Deployment:** The belt is placed around the waist, hips, or chest and secured using the **quick-release buckle**.
3. **Operational Synergy:** When worn, the back sleeve distributes weight and protects the skin, while the strap and "slip" mechanisms ensure all gear stays firmly in place during high-movement activities like concerts or sports.

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7. RISKS AND MISSING INFORMATION

Based on the current development stage of **itFits**, the project faces several strategic risks and lacks specific technical data required to move from a prototype phase to full industrial production.

1. Identified Risks

The sources highlight several critical threats that could impact the product's market viability:

- **Ethical and Cultural Perception:** Utilizing **real animal leather** for the back sleeve is a major cultural risk. While chosen for durability, younger, environmentally conscious consumers may view it as "murder," potentially alienating a core target demographic.
- **Fashion Volatility:** The current "bulky and flashy" aesthetic is trending now, but fashion indicators suggest a future shift back to **skinny belts**. If the trend changes before launch, the belt could become aesthetically obsolete for its primary fashion-focused audience.
- **Material Backlash:** Despite being recyclable, the heavy use of **polycarbonate plastic** may still draw negative criticism from "zero-plastic" advocates who do not view recycling as a sufficient environmental solution.
- **Technical Obsolescence:** Modular components like the **Zippo phone bag** run the risk of becoming obsolete whenever smartphone manufacturers change device dimensions.

2. Missing Measurements

Several physical and financial metrics are still marked as **"to be defined"** or require further precision:

- **Final Manufacturing Costs:** While prototype costs (straps at $279 MXN, leather at $400 MXN) are known, the **final industrial cost for injection-molded parts** remains undefined.
- **Industrial Weight Limits:** For the expansion into the blue-collar sector, the **maximum weight capacity** for the attachments needs specific measurement to ensure tools do not tear the slips.
- **Complete Module Catalog:** Dimensions for specialized modules beyond the Zippo v1 are missing, particularly for the "extreme version" variants. Also, missing the water bottle carrier in both versions.

3. Required Testing

The technical requirements table identifies several areas where **physical evidence** is still lacking:

- **Stress Testing:** The **UNBRK buckle** and injection-molded parts need documented stress test results to validate their "unbreakable" and "guaranteed resistance" branding.
- **Security Validation:** The efficacy of the belt's cross-body or chest positioning for **pickpocket prevention** in crowded festival scenarios has not yet been field-tested.
- **Bounce-Free Verification:** While conceptualized to prevent fatigue, formal **user testing in high-movement scenarios** (e.g., jumping at festivals or cycling) is needed to prove the "bounce-free" claim.
- **Durability Cycles:** Evidence is needed to confirm the **long-term reliability** of the memory foam and its ability to "instantly mold" to the body after thousands of uses.

4. Supplier and Logistical Confirmation

Several operational partnerships and logistical paths remain in the proposal stage:

- **Sourcing Verification:** Final verification is needed from partners like **Sutexsa** to confirm the 100% recycled origin of the nylon and polyester supply chain.
- **Artist Partnerships:** While artists like Jennie, David Guetta, and Anyma are named as ideal partners, **formal agreements** for special edition modules are still pending.
- **Recycling Logistics:** The **"Give Back" program** requires a confirmed logistics plan for how discarded modules will be collected, graded, and re-processed into new pellets.